

Date	8 July 2026	Project Name	Heart Disease Analysis
Team ID	—	Maximum Marks	2 Marks

PROJECT DESIGN PHASE

Proposed Solution Template

Team Leader: Mounika Sanapathi

Team Members: Thakasi Balaji, Kale Sai Praneeth, Navyasri Gonnuri, Shanthi Kumari Challa

The project team has filled in the structured parameters below to outline the technical design, deployment strategy, and clinical value propositions of the proposed analytics framework.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Medical centers and cardiologists gather substantial volumes of patient clinical data and cardiovascular metrics, but lack a structured, interactive framework to analyze it. Without explicit visibility into complex metric correlations (such as cholesterol levels, maximum heart rate achieved, and ST depression markers), patient risk categories, and clinical trends across different demographics, healthcare providers struggle to make timely, data-driven decisions regarding patient diagnostics, treatment priority, and preventive care strategies.
2.	Idea / Solution description	The proposed solution utilizes a comprehensive clinical heart disease dataset of approximately 50,000 records to build an interactive Tableau Dashboard and Tableau Story tracking cardiovascular risk factors, physiological metric correlations, patient demographic distributions, and diagnostic prevalence. These analytical visualizations are integrated into a Flask-based web application, providing doctors, cardiologists, and medical researchers with a single, highly accessible interface to explore business and clinical insights through interactive filters.
3.	Novelty / Uniqueness	Rather than relying on traditional static medical charts or tabular spreadsheets, the solution offers a dynamic, interactive Tableau Dashboard that lets users slice, filter, and explore patient records

S.No.	Parameter	Description
		natively. Combining this specialized Tableau Dashboard and sequential Story narrative within a robust Flask web framework creates a highly simple, user-friendly clinical intelligence interface that supports faster, safer, and data-driven diagnostic decision-making for healthcare networks.
4.	Social Impact / Customer Satisfaction	Cardiologists, healthcare practitioners, and medical research analysts gain a clearer, multi-dimensional understanding of patient cardiac risk profiles and physiological variations. This facilitates significantly quicker and more informed decisions regarding personalized clinical care planning, diagnostic prioritization, and preventive healthcare strategies. This leads to better-planned hospital resource deployment and, in turn, optimized patient recovery outcomes and diagnostic accuracy.
5.	Business Model (Revenue Model)	The solution can be offered via an institutional software-as-a-service (SaaS) subscription model for medical centers, hospitals, and private clinics, featuring tiered access to specialized diagnostic sub-modules, filtering tools, and automated report exports. Secondary value-driven revenue streams can come from custom dashboard integrations with existing Electronic Health Record (EHR) systems, custom medical research reporting extensions, and specialized diagnostic consulting engagements.
6.	Scalability of the Solution	The architecture is designed for full modular scalability, allowing it to easily expand by incorporating broader clinical datasets, longitudinal multi-visit histories, or multi-center institutional networks. The framework scales seamlessly to accommodate larger regional hospital chains and can host newly developed predictive analytics or AI-driven diagnostic risk models as medical requirements expand, all without requiring structural changes to the underlying dashboard layout or core application framework.